

Basics 22: Moments of the Kinetic Equation

III - Elementary plasma physics

How continuity, momentum, pressure, and energy equations emerge from phase-space dynamics

Textbook draft, version 1.0. Fluid plasma equations are velocity moments of a kinetic equation. This chapter performs the first three moment derivations carefully, showing where the pressure tensor, heat flux, collision terms, and closure problem enter. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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1 Learning objectives

After completing this note, the reader should be able to:

- Define velocity moments of a distribution function.
- Derive continuity from the kinetic equation.
- Derive momentum balance and the pressure tensor.
- Derive internal- and total-energy moment equations.
- Explain the infinite moment hierarchy.
- Distinguish conservative and convective fluid forms.

2 Prerequisites and reading strategy

- Basics 3, 5, 13, and 19.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Starting kinetic equation

For species s ,

$$\frac{\partial f_s}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f_s + \frac{q_s}{m_s} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \nabla_{\mathbf{v}} f_s = C_s[f] + S_s. \quad (3.1)$$

Assume f_s decays sufficiently rapidly as $|\mathbf{v}| \rightarrow \infty$ that velocity-space surface terms vanish. This technical condition is what permits integration by parts.

4 Zeroth moment: continuity

Integrate over velocity:

$$\frac{\partial}{\partial t} \int f_s d^3v + \nabla \cdot \int \mathbf{v} f_s d^3v + \frac{q_s}{m_s} \int (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \nabla_{\mathbf{v}} f_s d^3v = \int (C_s + S_s) d^3v. \quad (4.1)$$

The force term vanishes by velocity-space integration by parts because $\nabla_{\mathbf{v}} \cdot (\mathbf{E} + \mathbf{v} \times \mathbf{B}) = 0$. Thus

$$\boxed{\frac{\partial n_s}{\partial t} + \nabla \cdot (n_s \mathbf{u}_s) = S_{n,s}}. \quad (4.2)$$

Number-conserving collisions satisfy $\int C_s d^3v = 0$.

5 First moment: momentum

Multiply by $m_s \mathbf{v}$ and integrate. The force term becomes

$$q_s n_s (\mathbf{E} + \mathbf{u}_s \times \mathbf{B}). \quad (5.1)$$

Define the momentum-flux tensor

$$m_s \int \mathbf{v} \mathbf{v} f_s d^3v = m_s n_s \mathbf{u}_s \mathbf{u}_s + \mathbf{P}_s. \quad (5.2)$$

The conservative momentum equation is

$$\frac{\partial (m_s n_s \mathbf{u}_s)}{\partial t} + \nabla \cdot (m_s n_s \mathbf{u}_s \mathbf{u}_s + \mathbf{P}_s) = q_s n_s (\mathbf{E} + \mathbf{u}_s \times \mathbf{B}) + \mathbf{R}_s + \mathbf{M}_s. \quad (5.3)$$

Using continuity converts it to convective form.

6 Second moment: kinetic and internal energy

Multiply by $m_s v^2/2$ and integrate:

$$\frac{\partial \mathcal{E}_s}{\partial t} + \nabla \cdot \mathbf{F}_{\mathcal{E},s} = q_s n_s \mathbf{u}_s \cdot \mathbf{E} + Q_s. \quad (6.1)$$

The magnetic force performs no work. Decomposing $\mathbf{v} = \mathbf{u}_s + \mathbf{c}$ gives

$$\mathcal{E}_s = \frac{1}{2} m_s n_s u_s^2 + \mathcal{U}_s, \quad \mathcal{U}_s = \frac{1}{2} \text{tr} \mathbf{P}_s. \quad (6.2)$$

For an isotropic monatomic distribution, $\mathcal{U}_s = 3p_s/2$. The energy flux contains bulk transport, pressure work, and heat flux

$$\mathbf{q}_s = \frac{m_s}{2} \int c^2 \mathbf{c} f_s d^3v. \quad (6.3)$$

7 Pressure-tensor evolution and the hierarchy

Multiplying by $m_s \mathbf{c} \mathbf{c}$ produces a pressure-tensor equation involving the third central moment, while the heat-flux equation involves a fourth moment. Thus

$$\text{moment } k \longrightarrow \text{moment } k + 1. \quad (7.1)$$

No finite set closes itself without additional physics. Collisions may justify isotropic pressure and Fourier-like heat flow; collisionless magnetized plasmas often require anisotropic or Landau-fluid closures.

8 Conservative versus primitive formulations

A conservative equation has the form

$$\frac{\partial U}{\partial t} + \nabla \cdot \mathbf{F}(U) = S(U). \quad (8.1)$$

It naturally represents integral balances and shocks. Primitive equations evolve variables such as n , $\mathbf{u}$, and p and often expose physical interpretation. Numerical solvers may reconstruct primitive variables but update conserved variables to maintain correct balances.

9 Connection to kinetic MHD

Ideal MHD closes the bulk hierarchy with scalar pressure and an adiabatic law. Energetic-particle kinetic MHD instead keeps a distribution or selected kinetic moments for the fast population. Its pressure perturbation then feeds the fluid momentum or energy principle, retaining wave-particle resonances that scalar bulk closure cannot produce.

10 Computational laboratory

Represent a distribution on a velocity grid and evaluate density, momentum, pressure tensor, energy, and heat flux by quadrature. Verify the moments of a shifted Maxwellian and bi-Maxwellian. Demonstrate how velocity-domain truncation violates conservation.

11 Summary

- Fluid equations are weighted velocity integrals of the kinetic equation.
- Continuity, momentum, and energy correspond to the first three physically important moments.
- Pressure and heat flux encode unresolved velocity-space structure.
- The moment hierarchy is infinite and requires closure.
- Kinetic MHD preserves selected kinetic moments or distributions where fluid closure is inadequate.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
f_s	distribution function	Phase-space density of species s .
$\mathbf{P}_s$	pressure tensor	Second central moment.
$\mathbf{q}_s$	heat flux	Third central moment.
$\mathcal{E}_s$	total particle energy density	Bulk plus internal kinetic energy.
$C_s[f]$	collision operator	Velocity-space interaction term.
$S_{n,s}$	number source	Zeroth moment of external sources.

12 Exercises

1. Derive the zeroth moment and show why the Lorentz force contributes no particle source.
2. Derive the electromagnetic force in the first moment by integration by parts.
3. Decompose the momentum-flux tensor into bulk and pressure terms.
4. Show that magnetic forces do not appear in the kinetic-energy source.
5. Explain why a scalar equation of state is a closure rather than a consequence of the kinetic equation.

13 References

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